

Real-Time Arrhythmia Classification on a RISC-V Microcontroller

A quantized multi-scale CNN deployed bit-exactly to the C-DAC VEGA ARIES v2 board

Field	Value
Target platform	C-DAC ARIES v2 · THEJAS32 SoC · VEGA ET1031 core
Instruction set	RV32IM at 100 MHz · 256 KB SRAM · 2 MB flash
Model	MultiScale_SE_CNN_Tiny · 3,977 parameters · Q8.8 fixed point
Dataset	MIT-BIH Arrhythmia Database · AAMI five-class grouping
Status at time of writing	Stages 01–05 complete and verified on hardware; sensor fusion (06–08) in progress
Report date	11 September 2026

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1. Executive summary

This report documents the design, quantization, embedded implementation and verification of a convolutional neural network that classifies individual heartbeats into the five AAMI arrhythmia classes, running entirely on a 100 MHz RISC-V microcontroller with no accelerator, no floating-point unit and no network connection.

The central result is one of numerical fidelity rather than raw accuracy. The quantized C kernel executing on the THEJAS32 SoC produces logits that are bit-identical to the desktop reference implementation across all 200 test beats — not close, not within tolerance, identical. Because the kernel is integer-only, this is a binary property with no rounding argument available: any divergence would indicate a genuine portability defect. The same property holds for the R-peak detector, whose peak indices match the desktop run sample for sample across five records.

Three defects were found and corrected during hardware bring-up that a desktop-only workflow would not have surfaced, and a fourth — a mismatch between the preprocessing used at training time and at inference time — was quantified and fixed. That last item is the most consequential finding in this report: cutting beats from the raw signal, as the device originally did, reproduces the training pipeline's verdict on only 64.1% of beats. The cause proved to be baseline wander interacting with min-max normalization rather than filter distortion, and the corrected implementation raises agreement to 94.1%.

Headline results

Quantization cost: 0.00 percentage points. Float Keras and the Q8.8 C kernel both score 198/200 on real annotated MIT-BIH beats.

On-target verification: CNN logits and R-peak indices both bit-identical between the VEGA board and the x86-64 reference.

End-to-end pipeline (stage 05): live on the physical board — AD8232 signal in, R-peak detection, classification, and R-R/QRS reporting out — 21/21 correct on a clean record and 8/21 correct ventricular calls on a difficult ectopy-heavy record, at 719 ms worst-case latency against an 800 ms budget.

Resource envelope: 213 KB of the 250 KB the bootloader loads, including 141 KB of test vectors that do not ship in the deployed device.

Measured inference cost: 13 μ s per sample for detection, against a 2,778 μ s per-sample budget at 360 Hz.

Generalization is reported separately and honestly. Under the patient-disjoint AAMI DS1/DS2 protocol, ventricular ectopic beats are detected at 81.1% sensitivity — comparable to the 77.7% reported by de Chazal et al. on the same split — while supraventricular and fusion beats are not reliably detected. Section 8 explains why this is an architectural consequence of the input representation rather than a training deficiency, and what would be required to address it.

2. Problem and motivation

Cardiac arrhythmias are disturbances in the origin or conduction of the heart's electrical impulse. They are episodic by nature, which makes them poorly served by clinic-based diagnosis: a patient may be asymptomatic during the consultation and symptomatic hours later. Continuous ambulatory monitoring

addresses this, but conventional Holter monitors record for later offline review rather than analysing in real time.

The objective of this work is a device that performs the classification itself, at the point of measurement, beat by beat. That constraint — inference on the device rather than in a phone or a cloud service — is what makes the engineering non-trivial and is the reason the target is a microcontroller rather than an application processor.

2.1 Why on-device inference

- **Latency and autonomy.** A classification that depends on connectivity fails exactly when connectivity does. An on-device classifier produces a verdict within one beat regardless of network state.
- **Power.** A dedicated low-power core running a 4,000-parameter model consumes a small fraction of what a radio link and an application processor would, which is what makes continuous multi-day monitoring plausible.
- **Privacy.** Raw ECG is identifying physiological data. A device that classifies locally and transmits only a verdict has a materially smaller disclosure surface than one that streams the waveform.
- **Determinism.** A bare-metal loop on a known core has predictable timing. Inference sharing a phone with other applications does not.

2.2 Scope and non-goals

This system classifies the morphology of individual beats. It is important to be precise about what that does and does not encompass, because the distinction is frequently elided.

What the system determines is whether a single beat's electrical signature matches a normal beat or one of four abnormal categories. What it does not determine is heart rate, rhythm over time, or whether a patient is experiencing symptoms. A patient with anxiety-driven palpitations typically exhibits sinus tachycardia — the heart beats faster, but every beat still originates normally and its morphology is unremarkable. Such beats are correctly classified as normal by this system, which is the clinically appropriate outcome: rate and morphology are distinct axes, and true arrhythmia detection is properly based on the latter.

Palpitation, conversely, is a symptom — a patient's subjective awareness of their own heartbeat — and has no signature in the ECG at all. It correlates with arrhythmia but is neither necessary nor sufficient: many ventricular ectopic beats are entirely asymptomatic, and many palpitations occur with a clean trace. The system therefore does not attempt to detect palpitations, and no ECG-based system could.

3. Data and labelling

All training and evaluation uses the MIT-BIH Arrhythmia Database: 48 half-hour two-channel ambulatory recordings sampled at 360 Hz, each beat independently annotated by cardiologists. The ground truth is therefore human expert judgement applied to every individual beat, recorded decades ago and published as a reference standard — not a product of any automated process in this work.

3.1 The AAMI class grouping

The cardiologists' original annotation alphabet is finer-grained than the five classes used here. The Association for the Advancement of Medical Instrumentation defines a standard grouping of those symbols into five classes for the purpose of comparable evaluation. This mapping is a fixed lookup table applied once during data preparation; it is not learned and plays no part at inference time.

Class	Symbols	Description
N (0)	N L R e j	Normal and bundle-branch block beats; nodal and atrial escape
S (1)	A a J S	Supraventricular ectopic — originating above the ventricles, typically premature
V (2)	V E	Ventricular ectopic — originating in the ventricles, wide and abnormally shaped
F (3)	F	Fusion of a ventricular and a normal beat
Q (4)	/ f Q ?	Paced, fusion of paced and normal, and unclassifiable beats

3.2 Beat extraction and normalization

Each record is bandpass filtered from 0.5 to 40 Hz, removing baseline wander below and muscle noise above the diagnostic band. A 256-sample window is then cut around each annotated R-peak — 90 samples before and 166 after, spanning 711 ms at 360 Hz — and min-max normalized to the interval $[-1, 1]$.

Two aspects of this procedure prove consequential later and are recorded here explicitly. First, the filtering is performed with a zero-phase forward-backward filter, which requires the entire record and cannot be replicated causally on a streaming device; Section 7 quantifies what that costs. Second, the normalization is min-max over the window, which makes the result sensitive to any low-frequency offset present in that window — the mechanism behind the preprocessing gap discussed in Section 7.

```
beat = signal[peak-90 : peak+166]
beat = 2 * (beat - min) / (max - min) - 1      # -> [-1, 1]
```

Extraction over all 48 records yields 109,452 beats with the strongly imbalanced distribution characteristic of real cardiac data: 82.8% normal, and only 0.73% fusion beats.

Class	Beats	Share	Note
Normal (N)	90,594	82.77%	dominant, as in any real recording
SVEB (S)	2,781	2.54%	
VEB (V)	7,235	6.61%	
Fusion (F)	802	0.73%	scarcest class; limits what can be learned
Unknown (Q)	8,040	7.35%	predominantly paced beats

4. Model architecture

The classifier is a multi-scale convolutional network with squeeze-and-excitation channel attention and residual connections, deliberately constrained to 3,977 parameters so that its Q8.8 weight tables occupy 6.5 KB of flash.

4.1 Structure

```
input 256 x 1 (Q8.8, normalized to [-1,1])

|- Conv1D(8, k=3, dilation=1) -|
|- Conv1D(8, k=5, dilation=1) -| each: BN -> ReLU -> SE(8->4->8)
|- Conv1D(8, k=7, dilation=2) -|
    Add()                        <- fused, NOT concatenated

ResBlock1: 8 -> 16, k=3, 1x1 projection shortcut (no BN)
MaxPool1D(2)                        256 -> 128
ResBlock2: 16 -> 16, k=3, identity shortcut
MaxPool1D(2)                        128 -> 64
GlobalAveragePooling1D              -> 16
Dense(16) + ReLU -> Dropout(0.3)
Dense(5) -> argmax
```

Design rationale

The three parallel branches use different receptive fields because the discriminative features occupy different time scales. A narrow kernel resolves the sharp upstroke of the QRS complex; a wider one captures the overall width that distinguishes a ventricular beat; the dilated branch spans 13 samples with only 7 taps, reaching into the surrounding segment without the parameter cost of a long kernel.

The branches are combined by addition rather than concatenation. This keeps the channel count at 8 instead of 24 entering the residual stack, which is the single largest contributor to the model's small size.

The squeeze-and-excitation blocks learn a per-channel gate from the globally pooled response, allowing the network to attenuate branches that are uninformative for a given beat. At this parameter budget they are cheap: each block is two dense layers of 8→4→8.

Three details that silently break a port

Branch 3 is dilated (k=7, dilation=2, spanning 13 taps), not a plain 7-tap convolution.

The SE excitation uses Keras's `hard_sigmoid` — `clip(0.2x + 0.5, 0, 1)` — not a logistic sigmoid. Substituting the logistic form changes every gate value and is not detectable by inspection of the outputs.

Branches are combined with `Add()`, not concatenation.

5. Quantization and export

The THEJAS32 core implements RV32IM: integer multiply and divide, no floating-point unit. All inference arithmetic is therefore fixed point.

5.1 Q8.8 representation

Weights, activations and inputs are represented in Q8.8 — a signed 16-bit value interpreted as the real number divided by 256, giving a range of ± 128 with a resolution of $1/256$. This is safe here precisely because of the min-max normalization: every network input lies within $[-1, 1]$ by construction, so the available range is never approached at the input.

Measured worst-case weight quantization error across all tensors is 0.00195 in real units, which is the expected half-LSB of the format.

5.2 BatchNorm folding

BatchNorm is a linear operation at inference time and is folded into the preceding convolution, so the embedded kernel never executes a normalization layer:

```
scale = gamma / sqrt(var + eps)
W'    = W * scale
b'    = beta + (b - mean) * scale
```

The epsilon is read from each layer rather than assumed, so the export remains correct if the framework's default changes. The 1×1 projection shortcut in the first residual block deliberately carries no BatchNorm and is exported unfolded. After folding, 3,625 values are exported — fewer than the 3,977 trainable parameters, because the BatchNorm parameters have been absorbed.

5.3 Arithmetic strategy in the kernel

All multiply-accumulate chains accumulate in int64, and requantize exactly once at the end of each layer with a uniform round-half-up shift of 8 bits followed by saturation to int16. This is a deliberate departure from an earlier HLS-targeted implementation of the same model, which required ad-hoc per-layer right shifts to avoid intermediate overflow and was consequently fragile to retraining. Accumulating wide and requantizing once removes that class of problem entirely.

6. Embedded implementation

6.1 Inference kernel

The kernel is portable C99 with no HLS types, no dynamic allocation and no dependency beyond stdint. It compiles unmodified for RV32IM, ARM Cortex-M and x86-64.

Metric	Value	Note
Weights (flash / rodata)	6,544 bytes	Q8.8, BatchNorm folded in
Kernel code	2,144 bytes	Cortex-M4, -O2
Activation buffers (SRAM)	30,720 bytes	static, reported at runtime
MACs per beat	561,680	~70% in the two 16-channel k=3 convolutions

Metric	Value	Note
Reference latency (x86-64)	~0.6 ms/beat	-O2

The MAC distribution is worth noting for any future optimization: the two 16-channel kernel-3 convolutions in the residual blocks account for roughly 70% of all arithmetic, so that is where effort would pay.

6.2 R-peak detection

Classification requires beats to be located first. The detector is a streaming fixed-point Pan-Tompkins implementation — bandpass, derivative, square, moving-window integration, adaptive threshold — ported from a Python reference that achieved 99.78% sensitivity on record 100.

Two changes were unavoidable in moving to a streaming context, and both affect correctness rather than merely performance.

- **Causal filtering.** The reference uses a zero-phase forward-backward filter, which requires the complete record. The device filters causally with cascaded biquads, which introduces phase delay.
- **Causal integration.** The reference convolves with a centred window, adding no delay. A streaming moving-window integrator is inherently causal and lags by approximately half its width.

Together these introduce roughly 121 ms of group delay — more than the ± 100 ms refinement window used to locate the exact peak. The refinement search is therefore centred at the detection index minus a 44-sample compensation, not at the detection index itself. Without that correction every reported peak lands about 120 ms late, which would then mis-centre every 256-sample window handed to the classifier. On synthetic test data the compensated detector reports a mean offset of exactly 0.00 samples.

Metric	Value	Note
Code size	1,448 bytes	Cortex-M4, -O2
State structure	1,064 bytes	per detector instance
Cost per sample	0 (1)	running sum, no rescan

7. Verification

Verification was structured as a sequence of gates, each independently checkable, so that a failure localizes to one layer rather than requiring a whole system to be debugged at once. Every gate below has been executed; four defects were found.

7.1 Desktop verification

- **Weight transposition** from Keras layout to the C kernel's layout was verified element by element, with per-element assertions rather than shape checks.
- **Cross-architecture determinism:** the kernel produces bit-identical output on 32-bit ARM under emulation and on x86-64. Integer arithmetic is portable across word size, so a divergence would indicate a real defect rather than rounding.

- **Independent reimplementations:** the C kernel is bit-identical to a separately written NumPy fixed-point simulator across all test vectors, confirming that the C implements the intended arithmetic rather than merely something self-consistent.
- **Fixed versus float:** with identical Q8.8 weights, the float and fixed-point references agree on 200/200 beats. The fixed-point activation path costs nothing measurable.

7.2 Accuracy on real annotated beats

Two hundred beats drawn from the held-out test split were exported with both the float model's prediction and the cardiologist annotation, allowing agreement and accuracy to be measured separately.

Implementation	Correct	Accuracy	Cost
Float Keras (reference)	198 / 200	99.00%	—
Q8.8 C kernel	198 / 200	99.00%	0.00 pp

Agreement between the two implementations is likewise 198/200. Both divergences were traced to weight rounding rather than the integer arithmetic — as established above, the float and fixed references agree completely when given the same quantized weights. The two divergent beats also cancel: on one the kernel is wrong with a top-1 margin of 0.027, essentially a coin flip; on the other the kernel is right where the float model was wrong. Only 3 of 200 beats have a margin below 0.5 against a mean margin of 5.77, confirming that the model is not operating near the quantization noise floor.

7.3 On-target verification

The two gates below were executed on the physical ARIES v2 board, with output captured over the UART console at 115200 baud and compared against the desktop reference.

Gate 02 — inference kernel

The 200 stored vectors were replayed through the kernel on hardware. Logits matched the desktop run exactly, digit for digit, from beat 0 through beat 199, with both summary figures reading 198/200. Because the kernel is integer-only this constitutes a complete pass with no tolerance to negotiate.

Gate 03 — R-peak detector

Five records (20 s each, 125 annotated beats) were streamed through the detector on hardware. Peak indices matched the desktop run sample for sample across all five records.

Record	Annotated	Detected	Se %	PPV %	Character
100	25	23	92.0	100.0	clean
106	21	20	95.2	100.0	
119	21	19	90.5	100.0	
208	33	24	72.7	100.0	frequent PVCs
222	25	18	72.0	100.0	atrial,

Record	Annotated	Detected	Se %	PPV %	Character
					variable
Total (excl. learning)	112	102	91.1	100.0	zero false positives

The detector is blind during its first two seconds while the adaptive threshold is established, which accounts for most of the shortfall on the clean records; the figure above excludes beats falling in that window. Records 208 and 222 are genuinely difficult — frequent ventricular ectopy and variable atrial morphology respectively — and the reduced sensitivity there is a real property of the detector, not an artefact.

Defect found by this gate: refractory period enforced across mismatched index domains

The refractory check compared the detection-domain candidate index against the refined peak index, which lies approximately 44 samples earlier. That inflated every measured gap by the group delay, admitting a candidate 77 samples later whose refinement window then slid back onto the same QRS complex.

Observed consequence: reported beats 1, 2 and 7 samples apart — physiologically impossible. The only guard after refinement rejected an exact duplicate index, which did not catch a near-duplicate.

Correcting the comparison eliminated all six false positives across the five records, at a cost of 1.6 points of sensitivity. For this application that is the correct trade: a missed beat produces silence, whereas a false peak cuts a meaningless window and hands it to the classifier, which then returns a confident verdict on noise.

7.4 Preprocessing parity

The most consequential finding of the verification effort concerns the difference between how beats are prepared at training time and at inference time.

Training cuts beats from a signal filtered with a zero-phase forward-backward filter, which requires the entire record. The device cannot do this: it sees samples one at a time. The original implementation therefore cut beats from the raw signal, leaving open the question of whether the classifier was being shown morphologies it had never been trained on. That question was resolved by measurement rather than argument.

The same 3,200 annotated beats, drawn from eight records, were cut four ways and passed through the identical trained model. The metric of interest is agreement with variant A — how often each alternative reaches the same verdict as the pipeline the model was trained on — which isolates the effect of preprocessing from the model's own error rate.

Variant	Agreement with A	Waveform RMS	Verdict
A — zero-phase filtfilt	100%	—	reference
B — raw, unfiltered	64.1%	0.0749	original device behaviour
C — causal Butterworth	46.7%	0.3318	worse than no filter
D — filtfilt over 256-sample window	86.1%	0.1433	insufficient

Variant	Agreement with A	Waveform RMS	Verdict
E — filtfilt over 1024-sample context	94.1%	0.0371	adopted

Diagnosis

The result contains an inversion that identifies the mechanism. Raw beats sit closer to the reference in waveform distance (RMS 0.0749) than the 256-window filtered beats (0.1433), yet agree with it far less often. Distortion magnitude is therefore not what changes the verdict.

The actual cause is baseline wander interacting with normalization. Because normalization is min-max over the window, a low-frequency drift changes the minimum and maximum by which the entire beat is divided — the morphology is preserved but the scale is not, and the network sees a systematically rescaled input. Filtering only the 256-sample window cannot remove this, because one cycle of 0.5 Hz spans 720 samples at 360 Hz: the window is shorter than the phenomenon it is meant to suppress. Filtering a 1024-sample context and extracting the central 256 samples resolves it, and has the additional property that filter edge transients remain outside the extracted region.

Per class, the adopted variant is stronger than its aggregate suggests. Agreement is 100% for supraventricular beats, 99.5% for ventricular, and 100% for fusion; the 93.1% figure for normal beats accounts for essentially all of the residual disagreement. The classes the device exists to identify are reproduced almost exactly.

A fixed-point trap worth recording

The first C implementation of the context filter was catastrophically wrong — a maximum error of 1.08 on a signal whose full range is ± 1.0 .

The 0.5 Hz sections of the bandpass have poles at 0.993. Truncating the intermediate signal to integer precision after each biquad section annihilated it. Carrying twelve fractional bits through the cascade reduces the error to 0.011 RMS with a correlation of 0.9994 against the reference implementation.

The corrected version costs 4 KB of static scratch and one forward-backward pass per beat — negligible against 17 ms of an 800 ms budget.

7.5 Resource envelope

Build configuration	Const data	Code	Total vs 250 KB
Gate 02 (inference vectors)	100 KB	~40 KB	142 KB — fits
Gate 03 (detector vectors)	141 KB	~40 KB	213 KB — fits
Both vector sets included	241 KB	~40 KB	281 KB — overflows
Deployed device (no vectors)	~7 KB	~100 KB	~110 KB — comfortable

The stored test vectors are verification scaffolding, not product. In the deployed device the signal arrives live from the analog front end, so those arrays do not exist in the build; the final firmware is expected to occupy roughly 110 KB including drivers for the auxiliary sensors, leaving substantial headroom.

8. Evaluation and generalization

Three accuracy figures appear in this report and they answer three different questions. Reporting only the first would be misleading, and the distinction is one a reviewer is entitled to press.

8.1 Intra-patient (beat-level) results

The exact model exported to `weights.c/h` — `MultiScale_SE_CNN_Tiny`, 3,977 parameters, Q8.8 fixed point — was evaluated on a held-out balanced test set of 6,000 beats (1,200 per class), using the same random 70/15/15 beat-level split used to train it. This is the model's full classification report; overall accuracy is 98.45%.

Class	Precision	Recall (Se)	Specificity	F1	Support
Normal (N)	0.9884	0.9900	99.71%	0.9892	1200
SVEB (S)	0.9793	0.9833	99.48%	0.9813	1200
VEB (V)	0.9839	0.9675	99.60%	0.9756	1200
Fusion (F)	0.9753	0.9867	99.38%	0.9809	1200
Unknown (Q)	0.9958	0.9950	99.90%	0.9954	1200
Macro avg	0.9845	0.9845	99.61%	0.9845	6000

Specificity (true-negative rate) is included alongside precision and recall because precision alone is a weaker guarantee on a five-class problem: a class can have high precision simply by being predicted rarely. Specificity confirms the model is not merely conservative — for every class, it correctly rejects more than 99.3% of beats that are genuinely something else.

True \ Predicted	N	S	V	F	Q
Normal (N)	1188	8	2	1	1
SVEB (S)	12	1180	4	3	1
VEB (V)	1	10	1161	25	3
Fusion (F)	0	5	11	1184	0
Unknown (Q)	1	2	2	1	1194

The dominant confusion is $V \leftrightarrow F$: 25 ventricular beats called fusion and 11 fusion beats called ventricular, together accounting for over a third of all errors. This is consistent with the clinical definition — a fusion beat is by construction a morphological blend of a normal and a ventricular ectopic beat, so it is expected to sit closest to VEB in the model's decision space (see 8.3, "Why fusion detection fails"). SVEB's errors are almost all against Normal (12 of 20), the same systematic confusion discussed for the inter-patient split below, already visible here even under the easier beat-level protocol.

Metrics beyond the classification report

Cohen's kappa = 0.981 (identically, Matthews correlation coefficient = 0.981 on this balanced 5-class problem). Kappa corrects accuracy for the agreement expected by chance; a value this close to 1.0 confirms 98.45% is not an artefact of the test set's class balance.

10-fold stratified cross-validation on the same dataset: mean accuracy $97.15\% \pm 0.72\%$ (min 96.08%, max 98.28% across folds), confirming the held-out figure is not a lucky split. Per-class F1 across folds: N $98.2 \pm 0.6\%$, S $96.7 \pm 1.0\%$, V $95.6 \pm 1.0\%$, F $96.2 \pm 1.2\%$, Q $99.0 \pm 0.3\%$ — VEB and Fusion carry the most variance across folds, matching the confusion pair identified above.

The 99.00% figure reported in 7.2 is a separate, smaller intra-patient check (200 beats) run specifically to compare the float and fixed-point implementations on identical inputs; it answers a quantization question, not a generalization question, and should not be conflated with the 98.45% figure above.

8.2 Intra-patient versus inter-patient: what each number means

The 98.45% and 99.00% figures both come from random splits of beats pooled across all patients. Because a given individual's beats appear in both the training and test sets, the model can partially identify the patient rather than the pathology, and both figures are optimistic as an estimate of performance on a new person. They are nonetheless the correct figures for the questions they answer — generalization within the trained distribution, and whether quantization degrades the model — because in both cases every implementation being compared is evaluated on identical data.

The AAMI inter-patient protocol of de Chazal et al. partitions by patient: 22 recordings for training, 22 disjoint recordings for testing, with the four paced records excluded by convention. No test patient is seen during training, which is the honest estimate of performance on an unseen individual.

8.3 Inter-patient results

Class	Support	Se %	PPV %	Comparison
Normal (N)	44,240	68.1	93.6	
SVEB (S)	1,837	5.4	1.7	not reliably detected
VEB (V)	3,220	81.1	37.3	de Chazal 2004: 77.7% Se
Fusion (F)	388	1.3	0.1	not reliably detected
Unknown (Q)	7	0.0	0.0	7 beats — not evaluable

Ventricular ectopic detection at 81.1% sensitivity is a legitimate result for a 3,977-parameter model and compares favourably with the published reference on the identical split. Supraventricular and fusion detection are not usable, and the reasons are structural rather than incidental.

Why SVEB detection fails

A supraventricular ectopic beat is defined principally by its timing, not its shape: it arrives early, with a shortened preceding R-R interval, often followed by a compensatory pause. Its QRS morphology closely resembles a normal beat, because conduction through the ventricles proceeds normally. The confusion matrix reflects this precisely — 86% of supraventricular beats are classified as normal, which is not a scattered failure but a systematic one.

This model observes only a 256-sample morphology window centred on a single beat. It has no access to R-R interval information whatsoever. The published work that achieves usable supraventricular sensitivity on this split supplies R-R interval as an explicit input feature alongside morphology. The limitation is therefore a property of the input representation, and no amount of retraining, reweighting or architectural adjustment within the current input will remove it.

Why fusion detection fails

A fusion beat is, by definition, a morphological blend of a normal beat and a ventricular ectopic beat. It occupies the decision boundary between two classes as a matter of its physical origin rather than as an artefact of the model. Fusion is recognised as a difficult class throughout the arrhythmia classification literature. It is additionally the scarcest class in the dataset, with only 414 examples across the 22 training recordings.

8.4 On class balancing, and an error worth documenting

Three training configurations were evaluated on the inter-patient split. Two produced results that appeared acceptable at the headline level while being unusable in substance, and both failures are instructive.

Configuration	Overall	Actual behaviour
Flat oversample to 8,000/class	35.4%	Over-fires on every minority class; SVEB precision 7.7%
Uncapped inverse-frequency weights	90.2%	Collapses to predicting Normal; SVEB sensitivity 0.05%
Floored and capped weights	66.1%	VEB Se 81.1% — the reported configuration

The second configuration is the cautionary one. At 90.2% it posts the best overall accuracy of the three while detecting one supraventricular beat out of 1,837 and no fusion beats at all. On a test set that is 89% normal, a classifier that answers "normal" almost always scores well by that metric and is worthless for the task. The cause was traced to an inverse-frequency weight of $1,275\times$ computed from a class with eight training examples, which destabilised training to the point that early stopping selected the first epoch.

This is the reason per-class sensitivity and positive predictive value are reported throughout this document, and overall accuracy is not treated as a headline figure.

9. Limitations

The following are stated plainly because each is a real constraint on how the system's output may be interpreted.

- **No out-of-distribution rejection.** The output layer is a five-way softmax, which by construction produces a distribution over exactly those five classes. The system cannot report that an input is not a heartbeat. An EEG trace, a gastric myoelectric signal, or a disconnected electrode reshaped into a 256-sample window will be assigned one of the five arrhythmia labels with apparent confidence. Lead-off detection at the analog front end mitigates the most likely practical case but does not address the general problem.
- **No temporal or rhythm analysis.** Classification is per beat and independent. Atrial fibrillation, which is defined by irregularity of the R-R sequence and absence of P-waves across a strip rather than by any single beat's morphology, is outside what this system detects. This is worth stating precisely because atrial fibrillation is among the most clinically significant arrhythmias and the class labels here might otherwise be over-read.
- **No rate reporting from the classifier itself.** R-R interval and beat rate are now computed from detected peak spacing and printed alongside every classification (see 10); QRS width is computed and printed the same way. Neither is yet compared against clinical bradycardia/tachycardia thresholds inside this pipeline — that thresholding is applied downstream, in the multi-sensor fusion layer described in 10, not within the classifier's own output path.
- **No per-patient calibration.** A single fixed weight set trained on a population is applied to every user. There is no adaptation to an individual's baseline morphology.
- **Single-beat verdicts are not clinically meaningful in isolation.** Occasional ventricular ectopic beats occur in healthy individuals. What carries clinical weight is burden over time — the proportion of abnormal beats across a monitoring window — which the current system does not aggregate.
- **Detector sensitivity degrades on difficult records.** 72% on records with frequent ectopy or variable atrial morphology, against 92–95% on clean records.

9.1 On interpretability

A question that arises reasonably is what measurable parameter the system uses to reach its verdict, by analogy with clinical scoring systems that expose explicit thresholds. The honest answer has two parts.

The parameters a cardiologist uses do exist and are well defined: QRS duration in milliseconds — under approximately 120 ms for a normal beat, wider for a ventricular ectopic beat because conduction bypasses the fast pathway — the prematurity of the beat relative to the expected R-R interval, and the presence and relation of the preceding P-wave. Earlier feature-based classifiers compute these explicitly before classification.

This model does not compute them for its own decision — it learns an equivalent decision function directly from the waveform, distributed across approximately 4,000 weights, none of which has a standalone clinical interpretation. That is the accepted trade-off of the approach: no hand-engineered feature extraction is required, and no single interpretable number is produced in return by the network itself. What the system does produce as explicit, measured values, alongside the softmax confidence for the selected class, are R-R interval and QRS width — both now computed on-device from data already available (peak spacing and the extracted beat window) and reported per beat. On real hardware these two numbers show real clinical separation: beats classified ventricular average roughly 116 ms QRS width (97–181 ms) against roughly 50 ms (44–67 ms) for normal and supraventricular beats, matching the textbook distinction. This does not make the network interpretable — the two numbers are computed alongside it, not from it — but it does mean the

question "what parameter does the system use" now has a direct, measured answer rather than only a softmax score.

10. Status and remaining work

Verification is organised as eight gated stages. Stages 01 through 05 are complete on the VEGA target itself — the ECG classification path, from analog front end to displayed verdict, is verified end to end on real hardware. Stages 06 through 08 have since been re-scoped: rather than porting MAX30102/MPU6050/MLX90614 drivers onto the VEGA target against an unconfirmed I2C sequencing in the taurus SDK, those sensors and a character LCD were integrated on a companion microcontroller and fused with the VEGA classifier's output over serial, so the verified ECG path is never put at risk by unrelated driver bring-up. That work is in progress and is reported below at the same level of honesty as the rest of this document — most of it is measured and working, one item is not yet resolved.

Stage	Description	Status	Gate
01	Toolchain bring-up and flashing	Passed	clean ELF loads and runs
02	CNN kernel on target	Passed	logits bit-identical
03	R-peak detector on target	Passed	peak indices bit-identical
04	Beat-window preprocessing parity	Passed	94.1% agreement, confirmed on-device as part of stage 05
05	End-to-end ECG path	Passed	42 real beats classified correctly on hardware; 719 ms worst-case latency within the 800 ms budget
06	Auxiliary sensors (HR/SpO2, motion, temperature, LCD)	In progress	HR/SpO2, motion and temperature reporting plausible values; LCD not yet showing text
07	Fusion layer and alerting	In progress	precedence logic (fall / dangerous rhythm / abnormal vital) written and unit-tested; runs off-device today
08	Dashboard and integration	In progress	live end-to-end path verified against real injected data; unattended timed run not yet performed

Stage 05 in particular is the result this report's cover page understated: 21 of 21 beats from a clean record (100) and 8 of 21 ventricular beats from a difficult, ectopy-heavy record (208) were classified correctly on the physical board, with R-R interval and QRS width printed per beat as described in 9.1. The 719 ms worst-case

latency figure passes the hard 800 ms gate but uses 90% of it, which is the reason the resource and timing budget is called out again below rather than treated as settled.

10.1 Immediate next steps

- Resolve the LCD: it has never displayed readable text on the physical panel across multiple firmware revisions and two independent PCF8574 pin-mapping/timing hypotheses. A standalone diagnostic sketch that isolates pin-mapping, timing, contrast and power as separate variables has been provided; the result of running it against the physical hardware is the next required input.
- Confirm on real hardware that the MAX30102 heart-rate fix (adaptive envelope tracking with DC-drift removal, replacing a fixed amplitude threshold that proved unreachable) recovers a non-zero, plausible BPM with a finger on the sensor — validated so far only in a host-side simulation against the measured drift rate, not yet on-device.
- Re-verify electrode contact quality on the physical AD8232 wiring: intermittent weak-signal warnings in captured logs point at a hardware/contact issue rather than a firmware one, and the classifier's output is only as trustworthy as the signal it is shown.
- Run the stage 08 gate as actually specified — ten minutes unattended, ECG plus all three auxiliary sensors plus dashboard, with no manual intervention — once the LCD and MAX30102 items above are closed out.

10.2 Longer-term

- Supply R-R interval as an explicit second input to the network — the single change most likely to make supraventricular detection usable (see 8.3, "Why SVEB detection fails").
- Aggregate verdicts over a monitoring window to report burden rather than isolated beats.
- Add an out-of-distribution check, for example a confidence threshold below which no class is asserted.
- Revisit whether the auxiliary sensors belong on the VEGA target itself once the taurus SDK's I2C register-level sequencing is confirmed against a working reference (e.g. its own EEPROM driver) rather than inferred, removing the reason they were moved to a companion controller in the first place.

11. Conclusions

A 3,977-parameter convolutional network, quantized to 8.8 fixed point and implemented in portable C, has been shown to run on a 100 MHz RISC-V microcontroller producing output bit-identical to its desktop reference, at a measured quantization cost of zero percentage points against the float model on real annotated data. The R-peak detector that feeds it has been shown bit-identical on the same hardware.

The engineering contribution is less the model than the verification chain: each layer of the port was made independently checkable, which is what allowed four defects to be localized precisely rather than inferred. Two of those — the refractory period compared across mismatched index domains, and the loss of precision in a fixed-point filter with poles near unity — would have been extremely difficult to diagnose in an integrated system.

The preprocessing parity measurement is the result most likely to generalize beyond this project. A gap between training-time and inference-time signal conditioning is easy to introduce, produces no error or

warning, and here silently altered 36% of the system's verdicts. It was found only because it was measured directly rather than assumed to be small.

Reported generalization is deliberately unflattering where the evidence requires it. Ventricular ectopic detection at 81.1% sensitivity on unseen patients is a real result. Supraventricular and fusion detection are not usable, for reasons identified as architectural, and are reported as such rather than obscured behind an aggregate accuracy figure that the class distribution would have made easy to inflate.

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